GG-C-550B
April 25, 1977
SUPERSEDING
Fed. Spec. GG-C-550a
January 19, 1966

FEDERAL SPECIFICATION

COMPASS, DRAFTING, PIVOT, DROP SPRING BOW: PEN AND PENCIL, WITH CASE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers one type of drop-spring-bow drafting compass with pen and pencil attachments, and case.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-S-781 - Strapping, Steel, and Seals.
TT-I-528 - Ink, Drawing, Waterproof, Black.
UU-P-223 - Paper, Drawing.
CCC-C-531 - Cloth, Tracing.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.

FSC 6675

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PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation-Packaging, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society For Testing and Materials (ASTM) Standards:

B 122 - Copper-Nickel-Zinc Alloy (Nickel Silver) and Copper-Nickel Alloy Plate, Sheet, Strip, and Rolled Bar.
A 314 - Stainless and Heat-Resisting Steel Billets and Bars for Reforging.
D 2016 - Standard Methods of Test for Moisture Content of Wood.
E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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3. REQUIREMENTS

3.1 Description. The compass shall consist of a pivot rod, a pivot sleeve, and spring bow with interchangeable pen and pencil attachments. A case shall be furnished with each compass.

3.2 First article (first-produced compass with case). The supplier shall furnish one or more compasses with cases as specified (see 6.2) for examination and testing within in the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce a compass with case that will comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3.1 Copper-nickel-zinc alloy. The copper-nickel-zinc alloy shall conform to ASTM B 122, alloy 735 or 752.

3.3.2 Corrosion-resisting steel. Corrosion-resisting steel shall conform to ASTM A 314, type 302 or 430.

3.4 Design. The design of the compass shall be so as to enable the instrument to draw circles ranging in size from 1/32 inch to not less than 1 inch in diameter. Overall length of the compass shall be between 4 and 5 inches.

3.5 Pivot rod. The pivot rod shall be of corrosion-resisting steel and shall have a hardened needlepoint. The pivot rod shall have a knurled corrosion-resisting metal head, concave on the upper surface to fit the fingertip. The needlepoint shall have a 1/4-inch minimum range of movement above and below the point of the pen blades. The range of movement may be adjustable at the option of the supplier. When the pen is in contact with the tracing medium, the pivot rod shall extend at least 1/2 inch above the pivot sleeve.

3.6 Pivot sleeve. The pivot sleeve shall be of copper-nickel-zinc alloy, corrosion-resisting steel, chromium-plated brass, or aluminum. Aluminum, if used, shall be anodized. The pivot sleeve shall rotate freely and move vertically about the pivot rod with the minimum of

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tolerance. A knurled capscrew shall be fitted into the upper end of the pivot sleeve to prevent the pivot rod from falling out. The lower end of the pivot sleeve shall be tapered to provide a running fit about the pivot rod.

3.7 Spring bow. The spring bow shall be of corrosion-resisting spring steel. One end of the spring bow shall be securely attached to the pivot sleeve and the other end shall be attached to or have as an integral part a socket for the pen and pencil attachments.

3.8 Socket. The socket shall be of corrosion-resisting steel or copper-nickel-zinc alloy and shall be designed to accept and securely clamp the pen and pencil attachments.

3.9 Pen and pencil attachments. The pen and pencil attachments shall be designed with a shank which shall fit in one position into the socket and shall not turn after tightening of the socket setscrew.

3.9.1 Pencil. The pencil attachment shall be of corrosion-resisting steel or copper-nickel-zinc alloy and shall be designed to accept and securely clamp leads ranging in size from 0.073 to 0.083 inch in diameter.

3.9.2 Pen. The pen attachment shall be of corrosion-resisting steel, with or without tips of tungsten high-speed steel. The pen shall be either the spring-blade or spring-hinge type. An adjusting thumbscrew shall be provided. Pen blades shall be accurately set and evenly ground. The penpoints shall be hardened to a Rockwell hardness on the B scale of 84 to 86 for type 302 steel and 80 to 84 for type 430 steel. The hardness test point shall be between 1/4 and 5/16 of an inch back from the penpoint tip. The pen shall permit the drawing of clean, unbroken, and uniform lines, from 0.003 inch to 0.015 inch in width, without cutting into the drawing medium.

3.10 Screws. Adjusting thumbscrew heads shall be of copper-nickel-zinc alloy, corrosion-resisting steel, or aluminum, and shall have knurled edges. If made of aluminum, the screw heads shall be anodized. Screw studs and other screws shall be of corrosion-resisting steel. Threads shall be of a size and pitch to produce smooth action.

3.11 Case. The case shall be of wood or plastic with a hinged lid. The interior of the bottom of the case shall be made with recessions for the compass parts. The interior of the case shall be covered with a dark-colored velvet lining or similar material. Hardware used in

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construction of cases shall be of corrosion-resisting metal or shall be treated to resist corrosion. The case lining and outer covering, when used, shall be made fungus resistant by treating with a material which will not corrode the compass. When subjected to the accelerated aging test, the case shall not change appearance, shape, or dimensions, and the adherence of the covering or lining shall not be affected. When subjected to the impact test, the case shall not crack, break, chip, or otherwise be deleteriously affected.

3.11.1 Wood case. When applicable, each wood case shall be constructed of basswood, white pine, cedar, or other lightweight wood of clear stock. The wood shall have a moisture content of not less than 7 percent nor more than 14 percent when determined in accordance with ASTM D 2016.

3.11.1.1 Wood case covering. Unless otherwise specified (see 6.2), each wood case shall be covered with leather of good quality cattlehide or morocco finished in black.

3.11.2 Plastic case. When applicable, the plastic case shall be constructed of a nonflammable or slow-burning material of suitable hardness and rigidity. The plastic material shall have a low moisture absorption under severe humidity and temperature conditions.

3.12 Finish. All exposed surfaces of copper-nickel-zinc alloy shall have a finish produced by hard wheels. All exposed surfaces of steel shall be polished.

3.13 Identification marking (DoD only). The compass and case shall be identified in accordance with MIL-STD-130. In addition, each compass and case will have the manufacturer's name or trademark and the letters "U.S." stamped or otherwise permanently marked on the item.

3.14 Workmanship. Compasses and cases shall be free from any defects and imperfections which affect the appearance or which might impair the serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced compass and case inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First-produced compass and case inspection.

4.3.1 Examination. The first-produced compass and case shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first-produced compass and case shall be tested as specified in 4.5.2.1 and 4.5.2.2, respectively. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.5 percent defective for major defects and 4.0 percent defective for minor defects.

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4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1 and 4.5.2.2. AQL shall be 1.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The compass and case shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Dimensions not as specified.
103. Pivot sleeve does not move freely or rotate about the pivot point.
104. Socket does not accept and clamp the pen and pencil attachments.
105. Pencil attachment does not accept and clamp leads.
106. Adjusting screws do not operate smoothly.
107. Compass and compass parts do not fit the case properly.
108. Finishing not as specified.
109. Workmanship not as specified.

Minor

201. Identification marking not as specified.

4.5.2 Tests.

4.5.2.1 Compass. The compass shall be tested as specified in 4.5.2.1.1 and 4.5.2.1.2.

4.5.2.1.1 Performance. The test shall be performed using the pen attachment with the compass. Fill the pen with ink conforming to TT-I-528. Draw a number of circles with maximum and minimum diameters and with diameters ranging between the maximum and minimum on either drawing paper conforming to UU-P-223, type I, class 1, or tracing cloth conforming to CCC-C-531, type I. Check the diameter of the circles and the width and character of the drawn lines. Noncomformance to 3.4 and 3.9.2 shall constitute failure of this test.

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4.5.2.1.2 Hardness. The penpoints shall be tested in accordance with ASTM E 18. Nonconformance to the hardness specified in 3.9.2 shall constitute failure of this test.

4.5.2.2 Case. The case shall be tested as specified in 4.5.2.2.1 and 4.5.2.2.2.

4.5.2.2.1 Accelerated aging. The test shall be conducted in a temperature and humidity chamber. The case shall be subjected twice to each set of exposure conditions in the order specified in table I. After the second series of exposures, the case shall be examined. Any change in appearance, shape, or dimensions of the case or any change in adherence of the covering or lining shall constitute failure of this test.

TABLE I. Accelerated aging test

Exposure	Length of time (Hours)	Temperature (°C)	Relative humidity (Percent)
First	24	60	85 to 90
Second	24	60	Less than 10
Third	24	60	85 to 90
Fourth	24	0	Approx. 100

4.5.2.2.2 Impact. Place the compass and parts in the case. Drop the case from a height of not less than 5 feet onto a concrete platform. Any evidence of cracks, breaks, or chips on the case shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

110.	Materials, methods, and containers not as specified. Each incorrect material, container, or method shall constitute one defect.
111.	Consolidation quantities not as specified for level A.
112.	Fiberboard boxes not closed as specified for level A or B.
113.	Strapping not as specified for level A.
114.	Marking illegible, incomplete, or incorrect for level A or C.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, level B, or Commercial, as specified (see 6.2).

5.1.1 Level A. Each case, containing the compass and attachments, shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.1.1 Consolidation. The preserved compasses, in quantities of 10, shall be placed in close-fitting boxes conforming to PPP-B-636, W5c, style optional. The boxes shall be closed in accordance with method V of the appendix to the box specification.

5.1.2 Level B. Each case, containing the compass and attachments, shall be preserved in accordance with MIL-P-116, method III.

5.1.2.1 Consolidation. Consolidation shall be as specified in 5.1.1.1 for level A. When specified (see 6.2), boxes shall be class domestic, variety SW, grade 200, and closure shall be as specified for class domestic boxes.

5.1.3 Commercial. The compasses and attachments, contained in their cases, shall be preserved to afford protection against deterioration and damage during shipment from the contractor to the initial destination.

5.2 Packing. Packing shall be level A, level B, or Commercial, as specified (see 6.2).

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5.2.1 Level A. The compasses, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, grade B, or PPP-B-621, Class 2. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.2 Level B. The compasses, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-636, V3c, style optional, or PPP-B-640, class 2, grade A, style optional. The gross weight or size shall not exceed the limitation of the box specification. The boxes shall be waterproof closed in accordance with the appendix to the box specification.

5.2.3 Commercial. The compasses, preserved as specified in 5.1, shall be packed in containers to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be by any means which provides legibility and shall include the national stock number (NSN) or part number, nomenclature, quantity and unit of issue, contract, purchase order, or delivery order number, and address.

5.3.3 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. The drop-spring-bow compass is intended to be used to draw small circles in general drafting practice.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first-produced model, and number of compasses and cases required (see 3.2).

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(c) If case covering material is to be other than leather, specify material required (see 3.11.1.1).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(e) When boxes shall be class domestic (see 5.1.2.1).
(f) When finish A strapping is required (see 5.2.1).

6.3 First-produced compass and case. Any changes or deviations of production compasses and cases from the first-produced compasses and cases during production will be subject to the approval of the contracting officer. Approval of the first-produced compasses and cases will not relieve the supplier of his obligation to furnish compasses and cases conforming to this specification.

Preparing activity: Army - ME

MILITARY INTERESTS:
Custodian:
Army - ME

Review activity:
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA - FSS
COM - NBS
VA - DMS

Project 6675-0637

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DOCUMENT IDENTIFIER AND TITLE GG-C-550B Compass, Drafting, Pivot, Drop Spring Bow: Pen and Pencil, with Case

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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